

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 3**

**COURSE NAME: INTERNET PROGRAMMING**  
**COURSE CODE: CSA4305**

### **COURSE OUTCOME COVERED**

**CO-1:** Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

**Weightage: 10%**

**QUESTIONS: Total Marks: 10**

### **Scenario: Web Development Standards Implementation**

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.

The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

### **Data Snippets Provided**

#### **Snippet 1: Video Embedding Options**

A. `<media src="video.mp4"></media>`

- B. <video src="video.mp4" controls></video>
- C. <movie src="video.mp4"></movie>
- D. <vid src="video.mp4"></vid>

### Snippet 2: Form Submission Methods

- A. GET
- B. POST
- C. PUT
- D. DELETE

### Snippet 3: Navigation Elements

- A. <div>
- B. <nav>
- C. <section>
- D. <header>

### Snippet 4: XML Syntax

- A. <note><to>User</to><from>Admin</from>
- B. <note><to>User</to><from>Admin</from></note>
- C. <note><to>User</from></note>
- D. <note to="User"></note>

### Snippet 5: Web Communication Protocols

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

### Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

### **Code of Conduct:**

*I M.RESHWANTH(192411235) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student: M.RESHWANTH**

### RUBRICS FOR CALCULATION:

| Criteria                | Weightage | Level 4 – Exemplary                                              | Level 3 – Proficient                           | Level 2 – Developing                     | Level 1 – Beginning                                  |
|-------------------------|-----------|------------------------------------------------------------------|------------------------------------------------|------------------------------------------|------------------------------------------------------|
| Understanding of Case   | 25        | Thorough understanding ; clearly identifies all key issues       | Good understanding with most issues identified | Basic understanding ; some issues missed | Poor understanding; problem not identified correctly |
| Application of Concepts | 25        | Effectively applies relevant concepts to analyze the case deeply | Applies appropriate concepts with minor gaps   | Limited application of concepts          | Fails to apply relevant concepts                     |
| Analysis                | 20        | In-depth analysis with strong reasoning and insights             | Good analysis with logical reasoning           | Basic analysis with limited depth        | Weak or no analysis; lacks reasoning                 |
| Solution Design         | 20        | Innovative, feasible solutions with strong justification         | Logical solutions with adequate justification  | Basic solutions with weak justification  | Incorrect or no solution provided                    |
| Clarity                 | 10        | Clear, well structured, and easy to understand                   | Organized and understandable presentation      | Limited clarity and structure            | Poor clarity; difficult to understand                |

# CSA4305-Internet Programming

M.RESHWANTH(192411235)

## Question 1: Video Embedding

Options:

- A. `<media src="video.mp4"></media>`
- B. `<video src="video.mp4" controls></video>`
- C. `<movie src="video.mp4"></movie>`
- D. `<vid src="video.mp4"></vid>`

**Correct Option: B** — `<video src="video.mp4" controls></video>`

**Reason for Selection**

- The `<video>` element is the standard HTML5 tag used to display videos on a web page.
- It is supported by all modern web browsers.
- The `controls` attribute provides built-in play, pause, volume, and fullscreen controls.
- It allows users to watch videos without installing additional software or plugins.

**Explanation of Other Options**

- A. `<media>` – This is not a valid HTML5 element for displaying videos.
- C. `<movie>` – This tag is not defined in the HTML specification.
- D. `<vid>` – This is not a recognized HTML element and browsers do not support it.

## Question 2: Form Submission Method

Options:

- A. GET
- B. POST
- C. PUT
- D. DELETE

**Correct Option: B** — POST

**Reason for Selection**

- POST sends the form data in the HTTP request body instead of the URL.

- It is more secure because sensitive information is not displayed in the browser address bar.
- It is suitable for registration forms, login pages, and file uploads.
- POST can transmit large amounts of data.

### Explanation of Other Options

- **A. GET** – Sends data in the URL, making sensitive information visible.
- **C. PUT** – Used to update or replace an existing resource, not for HTML form submission.
- **D. DELETE** – Used to remove resources from the server and is not suitable for registration forms.

## Question 3: Navigation Element

### Options:

- A. `<div>`
- B. `<nav>`
- C. `<section>`
- D. `<header>`

**Correct Option: B — `<nav>`**

### Reason for Selection

- `<nav>` is the HTML5 semantic element used to define navigation links.
- It helps search engines understand the website structure.
- It improves accessibility for screen readers.
- It makes the webpage more organized and standards-compliant.

### Explanation of Other Options

- **A. `<div>`** – A generic container that does not provide semantic meaning.
  - **C. `<section>`** – Used for grouping related content, not specifically for navigation.
  - **D. `<header>`** – Represents introductory content and may contain a `<nav>` element, but it is not specifically meant for navigation.
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## Question 4: XML Syntax

### Options:

- A. `<note><to>User</to><from>Admin</from>`
- B. `<note><to>User</to><from>Admin</from></note>`
- C. `<note><to>User</from></note>`

- D. `<note to="User"></note>`

**Correct Option: B** — `<note><to>User</to><from>Admin</from></note>`

### Reason for Selection

- Every opening XML tag has a matching closing tag.
- The tags are properly nested.
- `<note>` acts as the root element.
- The XML document follows well-formed XML rules.

### Explanation of Other Options

- A. Missing the closing `</note>` tag.
- C. The opening and closing tags do not match (`<to>` is closed with `</from>`).
- D. The closing tag is incomplete, making the XML invalid

## Question 5: Web Communication Protocol

### Options:

- A. FTP
- B. HTTP
- C. SMTP
- D. SNMP

**Correct Option: B** — HTTP

### Reason for Selection

- HTTP (HyperText Transfer Protocol) is the standard protocol used for communication between web browsers and web servers.
- It transfers web pages, images, videos, and other web resources.
- It supports request and response communication.
- HTTPS is the secure version of HTTP used in modern web applications.

### Explanation of Other Options

- A. **FTP** – Used for transferring files between computers, not for loading web pages.
- C. **SMTP** – Used for sending emails between mail servers.
- D. **SNMP** – Used for monitoring and managing network devices, not for web communication.